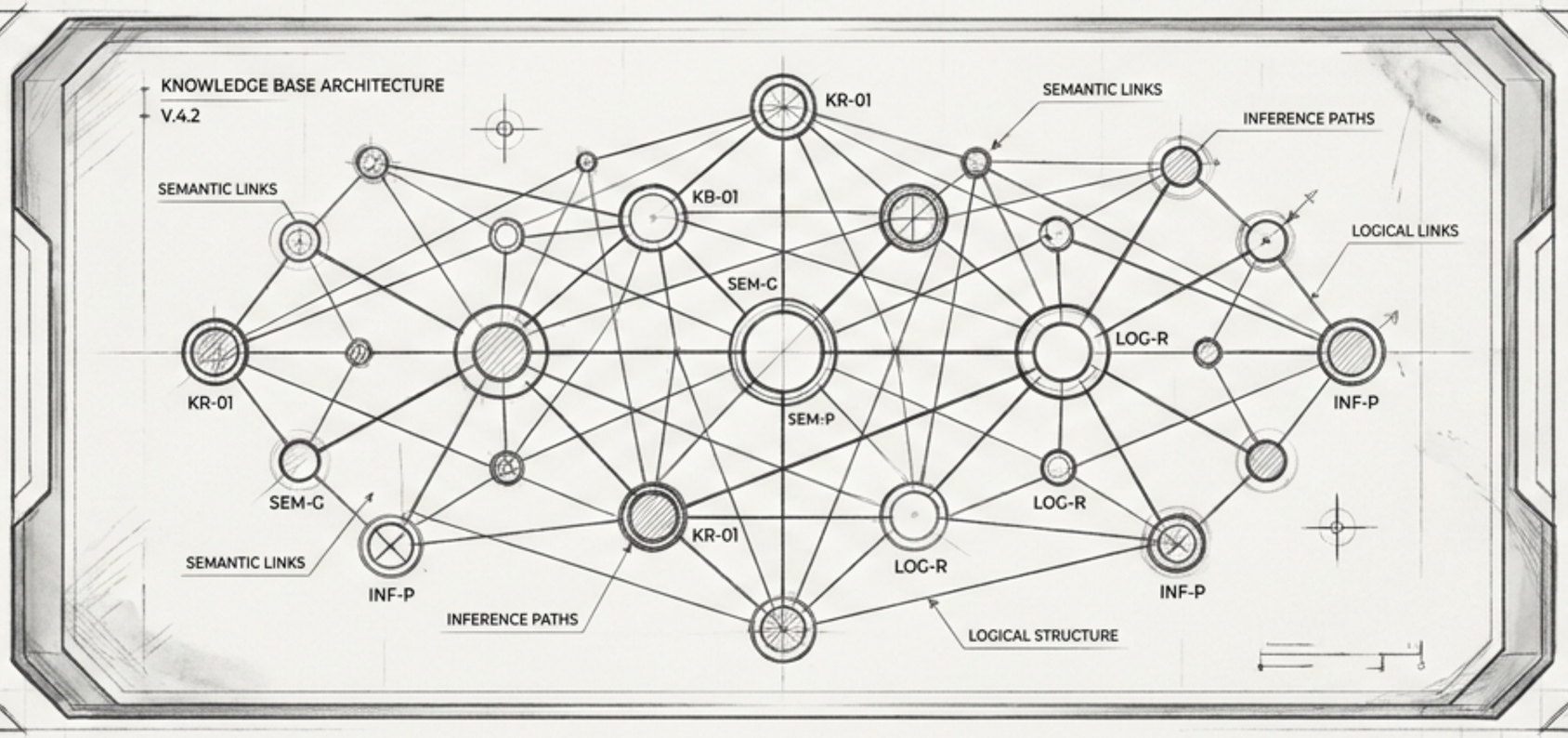
DIAGNOSTIC ASSESSMENT IN KNOWLEDGE REPRESENTATION AND REASONING

A SOCRATIC MASTERCLASS & EVALUATION



Terminal active. 15 diagnostic scenarios loaded.
Select optimal parameters to proceed.

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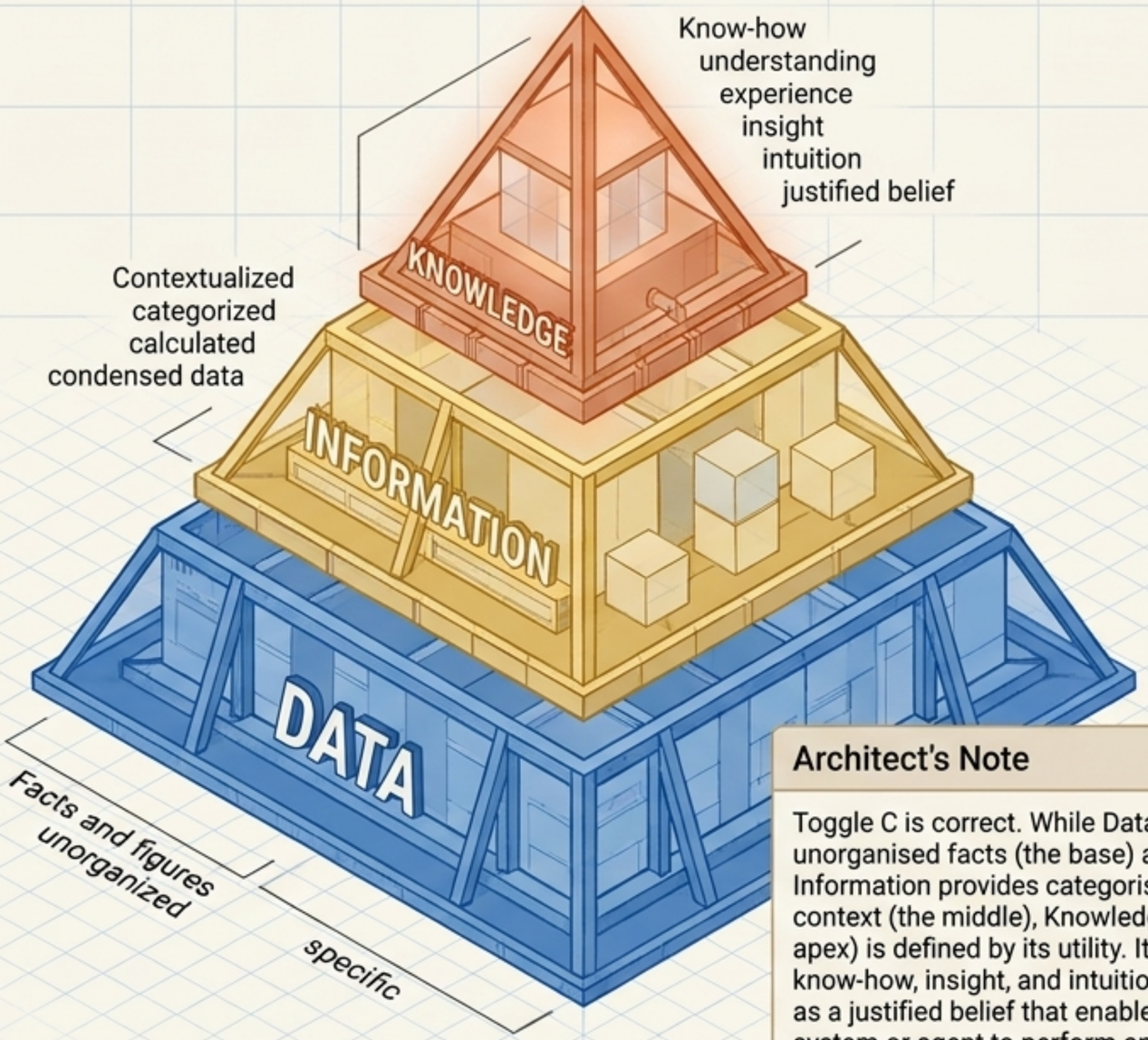
Diagnostic Prompt 01

In the context of AI modelling, which definition accurately captures the critical threshold where 'information' transforms into 'knowledge'?

- **A.** Categorised and condensed facts relaying specific figures.
- **B.** The formal syntax used to arrange alphanumeric symbols in a database.
- **C. A justified belief** that increases an entity's capacity for effective action.
- **D.** Universally true data perceived through sensors before interpretation.



Architect's Note

Toggle C is correct. While Data represents unorganised facts (the base) and Information provides categorised context (the middle), Knowledge (the apex) is defined by its utility. It includes know-how, insight, and intuition—acting as a justified belief that enables a system or agent to perform an effective action.

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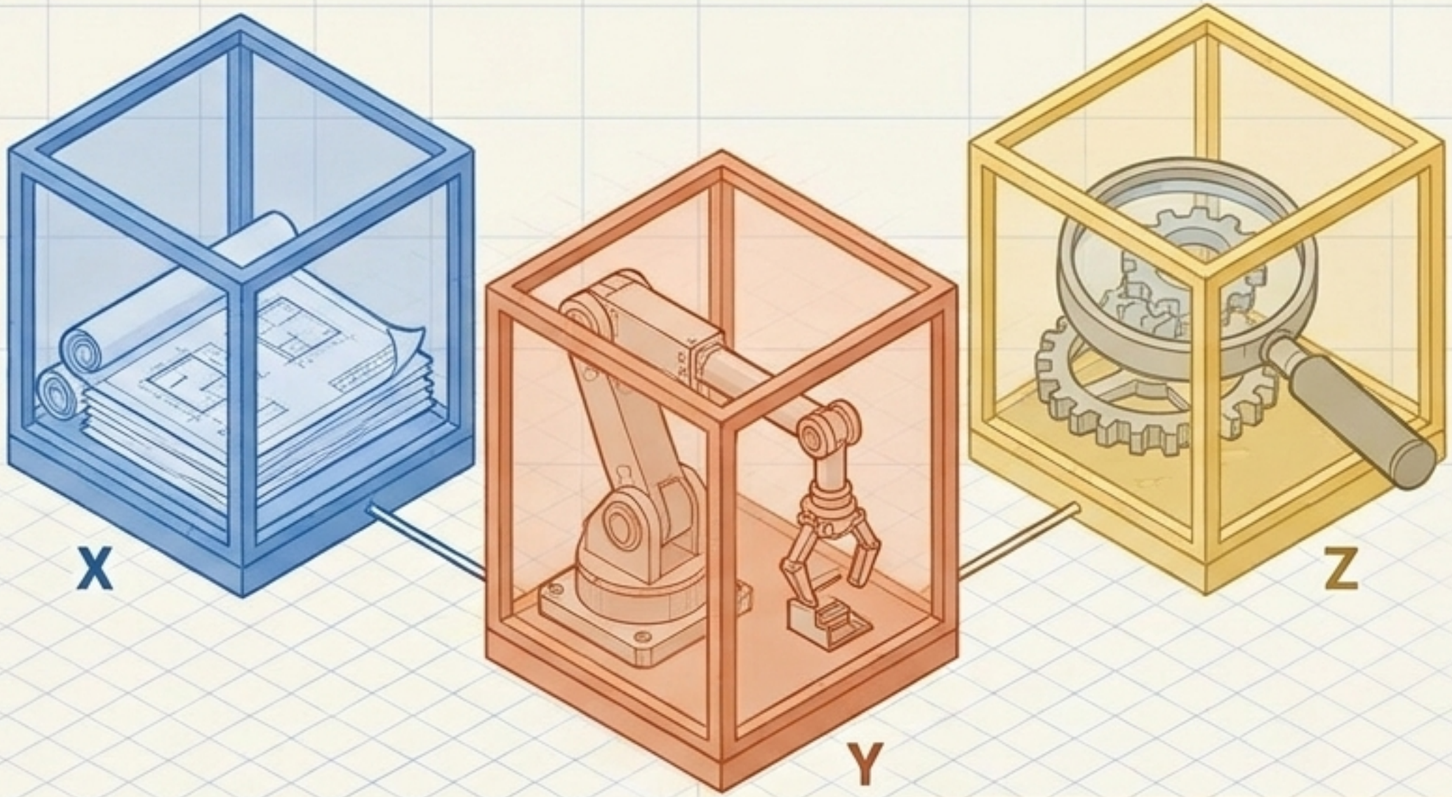
Diagnostic Prompt 02

An AI engineering firm builds a factory system. Module X stores component specifications. Module Y contains the assembly algorithms. Module Z evaluates whether Module Y's algorithms are efficient enough to run. How are these knowledge types classified?

Structural toggles

- **A.** X: A priori,
  Y: A posteriori,
  Z: Tacit
- **B. X: Declarative,**
  Y: Procedural,
  Z: Meta-knowledge
- **C.** X: Relational,
  Y: Conditional,
  Z: Structural
- **D.** X: Procedural,
  Y: Declarative,
  Z: A priori



Architect's Note

Toggle B is correct. Module X contains 'Know-about' (facts and concepts), which is Declarative. Module Y contains 'Know-how' (rules and procedures), which is Procedural. Module Z evaluates the system's own knowledge rules, making it Meta-knowledge (knowledge about knowledge).

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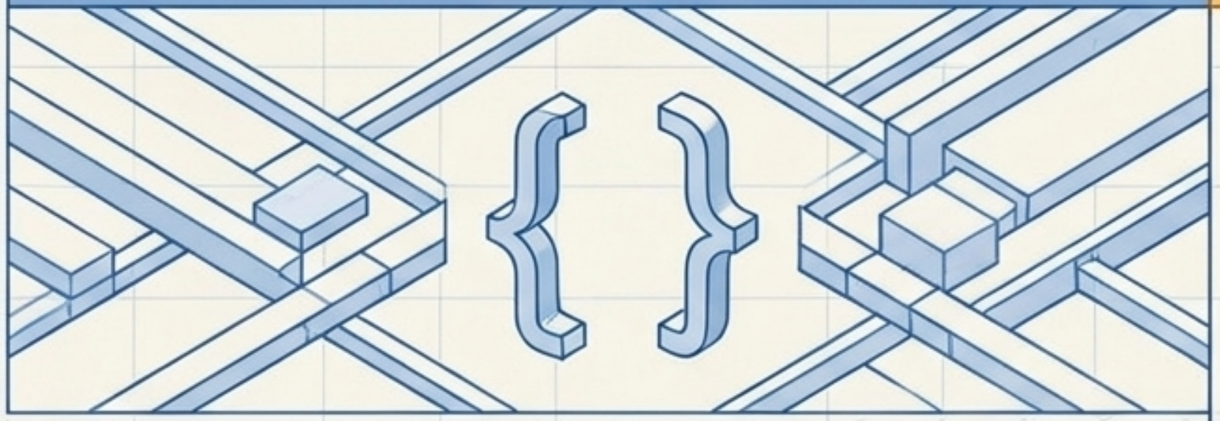
Syntactic Rules



Semantic World

Diagnostic Prompt 03

A database system rejects the input string: marital_status(025) is male. What specific foundational rule of a Knowledge Representation Language has been violated?

- **A. Incorrect Semantics**: The sentence makes a false statement about the world.
- **B. Incorrect Syntax**: The arrangement of atomic tokens is illegal within the alphabet of symbols.
- **C. Contextual Ambiguity**: The language requires natural language descriptors.
- **D. Inferential Inadequacy**: The system cannot derive new knowledge from this fact.

Architect's Note

Toggle B is correct. Syntax governs the legal arrangement of symbols. Using "025" as a name parameter violates the structural rules (syntax) of this database language. Conversely, 'marital_status(John is dlvorced' (if he is actually single) would be syntactically correct but semantically incorrect, as Semantics dictates what the arrangement means in the real world.

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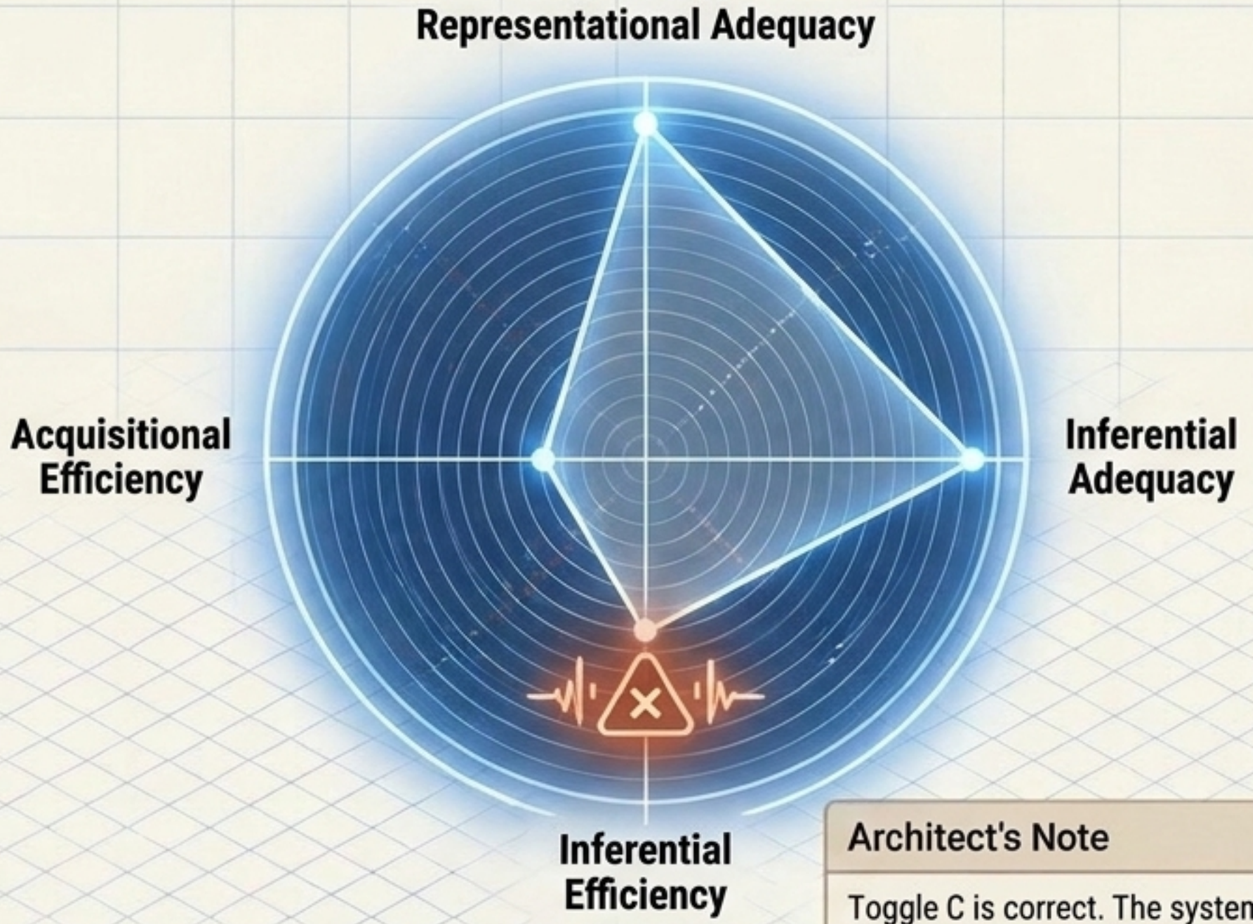
Diagnostic Prompt 04

An autonomous vehicle's AI successfully stores all map entities and can logically deduce new collision trajectories. However, it takes 4 minutes to calculate a braking decision, resulting in a crash. Which theoretical requirement is catastrophically failing?

- A. Representational Adequacy
- B. Inferential Adequacy
- C. Inferential Efficiency
- D. Acquisitional Efficiency



Architect's Note

Toggle C is correct. The system possesses Representational Adequacy (it can model the needed concepts) and Inferential Adequacy (it can derive the new knowledge). However, it lacks Inferential Efficiency: the ability to direct the inferential mechanism productively and respond within limited resource constraints (in this case, computational time).

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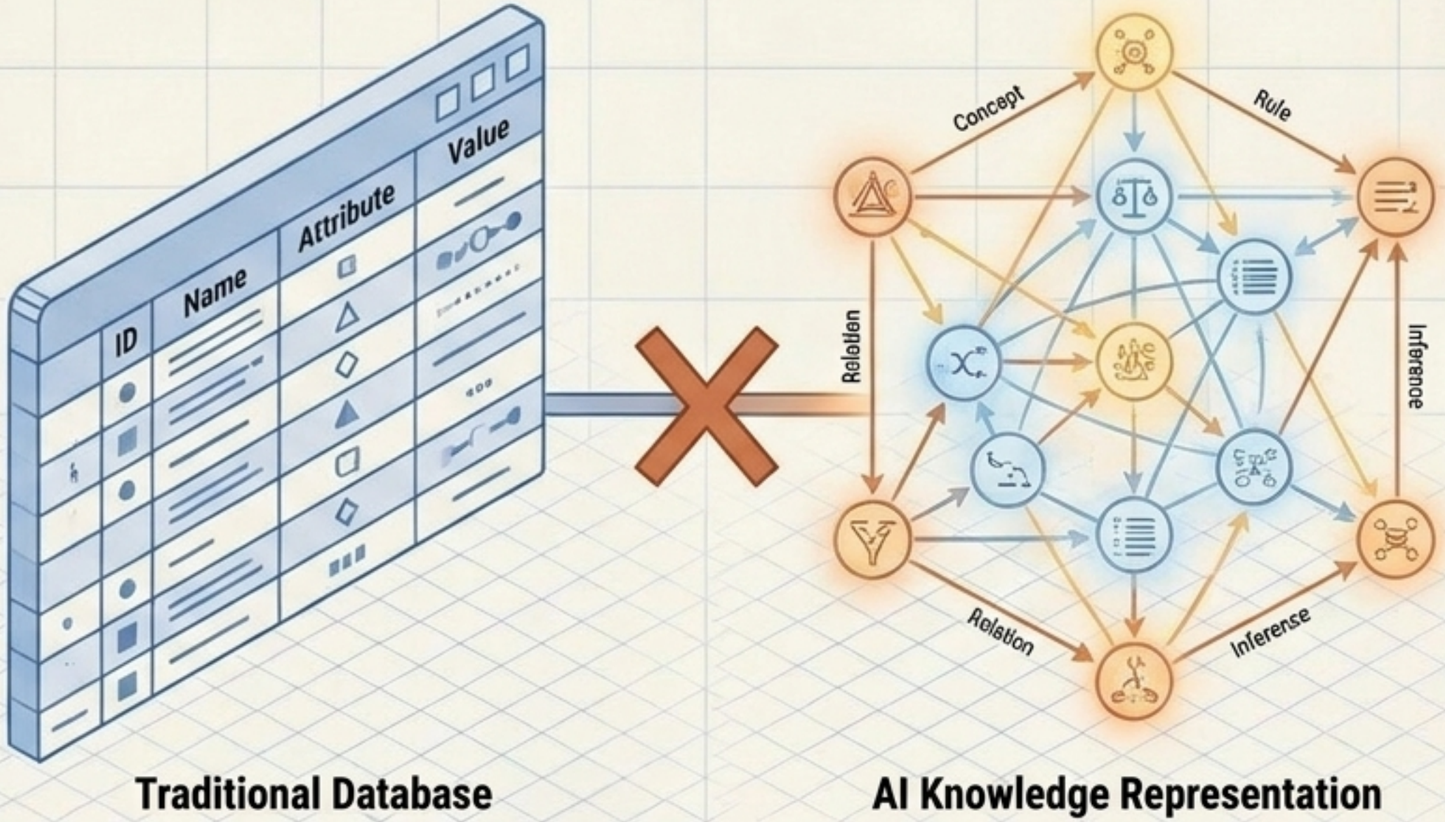
Diagnostic Prompt 05

Despite their immense ability to efficiently store and process large amounts of data, why are traditional Database Systems ultimately limited as a general-purpose AI Knowledge Representation scheme?

- **A.** Their implementation is neither concise nor fast enough for AI.
- **B.** They cannot represent entities or relationships between entities.
- **C.** Their syntax and semantics are too ambiguous and context-dependent.
- **D.** Reasoning is fundamentally restricted to simple lookup operations.



Traditional Database **AI Knowledge Representation**

Architect's Note

Toggle D is correct. While simple databases are effective, computable, and can clearly represent entities and relationships, the only reasoning inherently possible is simple lookup. For advanced AI, we require sophisticated processing mechanisms that can infer new, unstated knowledge. (Note: Natural Language suffers from the ambiguity mentioned in Option C).

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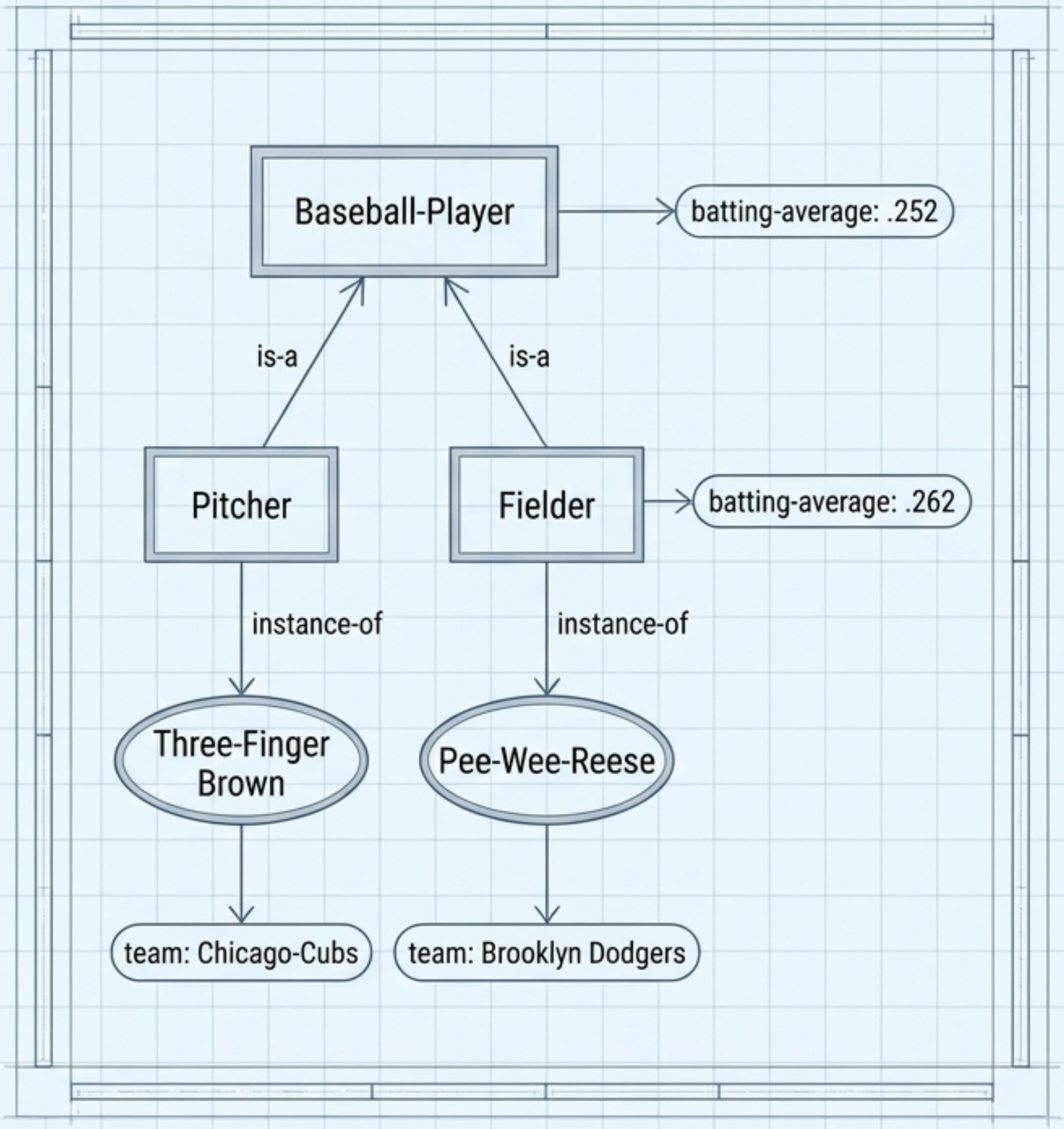
Diagnostic Prompt 06

Examine the Semantic Network mapping baseball players. By tracing the links, what is the 'team' value for **Pee-Wee-Reese**, and what is his 'batting-average'?

- A. Team: Chicago-Cubs; Average: .252
- B. Team: Brooklyn Dodgers; Average: .262
- C. Team: Brooklyn Dodgers; Average: .252
- D. Team: Unknown; Average: .106

Architect's Note

Toggle B is correct. Pee-Wee-Reese has a direct link for 'team' pointing to Brooklyn Dodgers. He is an instance of a 'Fielder', so he inherits the 'batting-average' of .262 directly from the Fielder node, overriding the more general Baseball-Player average of .252.



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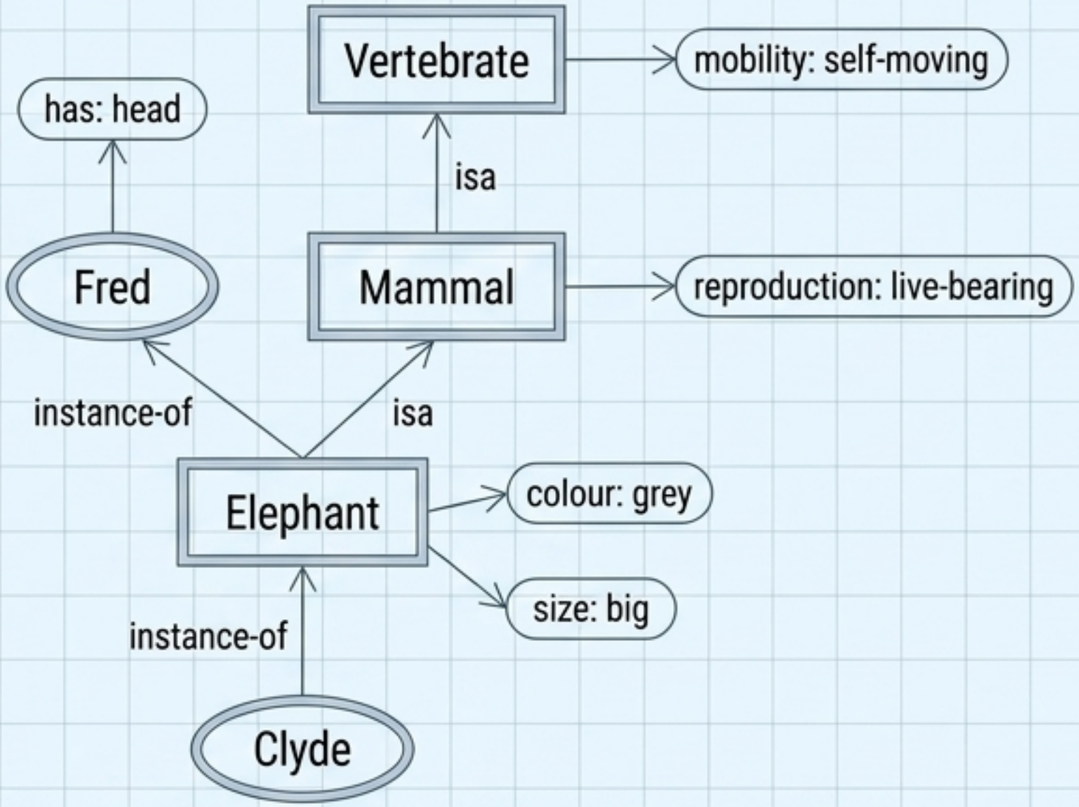
Diagnostic Prompt 07

Analyse the vertebrate inheritance chain. Which combination of properties does the instance ‘Clyde’ definitively inherit through ‘isa’ relationships?

Properties

Anayse the vertebrate inheritance chain. Which the instance “Clyde” definitively inherit through ‘isa’ relationships?

- [ ] A. Colour: grey; Habitat: ground; Legs: 4
- [ ] B. Size: big; Proboscis: yes; Head: yes
- [x] C. Reproduction: live-bearing; Mobility: self-moving
- [ ] D. Instance: Fred; Has: head



Architect’s Note

Toggle C is correct. While Clyde is grey and big (direct traits of the elephant node), the prompt specifically asks for traits inherited via ‘isa’ chains. Clyde ‘isa’ elephant, which ‘isa’ mammal (reproduction: live-bearing), which ‘isa’ vertebrate (mobility: self-moving).

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FRAME-BASED KNOWLEDGE REPRESENTATION

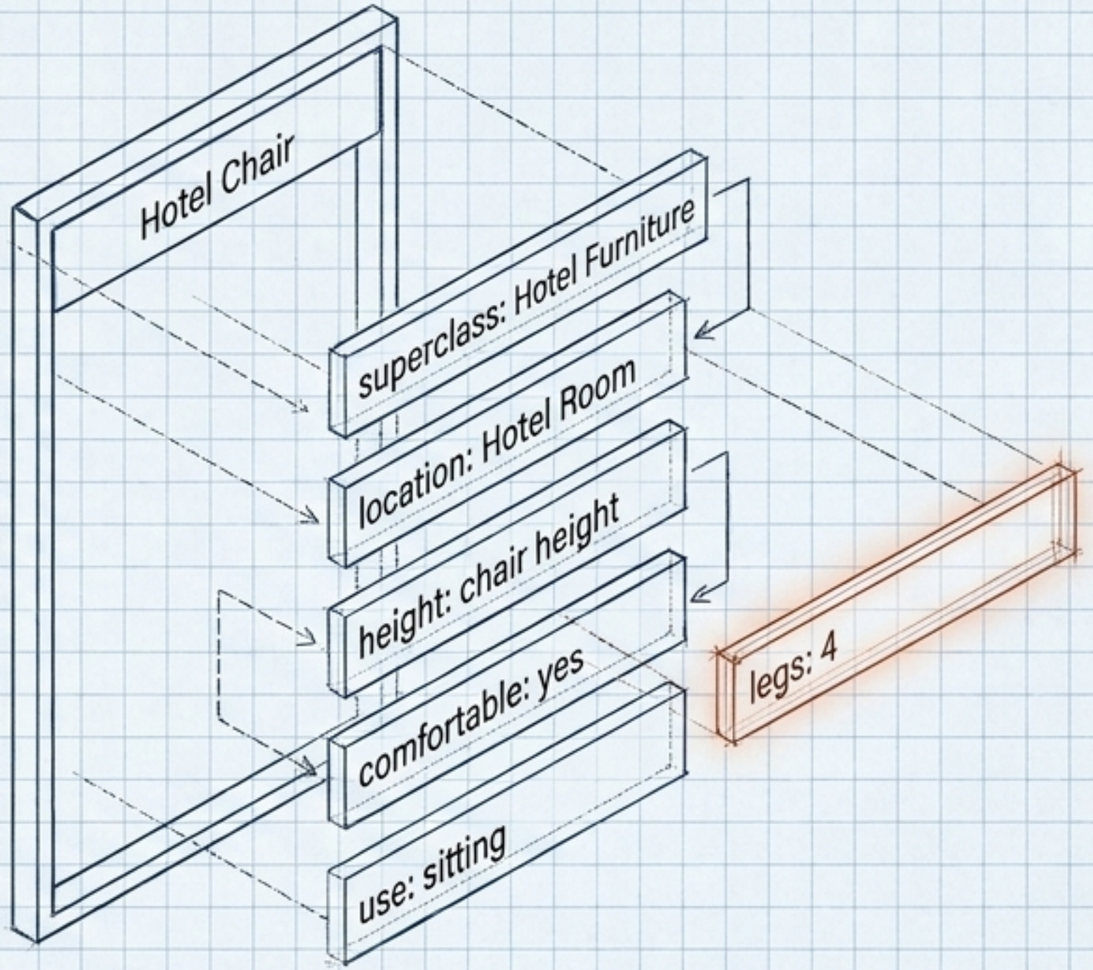
Diagnostic Prompt 08

In a Frame-based Knowledge Representation system, what specific mechanism allows the system to assume a hotel chair has 4 legs without requiring explicit input for every new chair instance?

- ○ A. Procedural information attachment
- ◉ B. Frame default information in a slot
- ○ C. Semantic ambiguity resolution
- ○ D. Frame identification pointers

Architect's Note

Toggle B is correct. Frame default information provides slot values (like chairs having four legs) that are taken to be true unless evidence to the contrary is found. This enables representational adequacy and suppresses irrelevant details during initial object creation.



EXPLODED VIEW - HOTEL CHAIR FRAME ANATOMY

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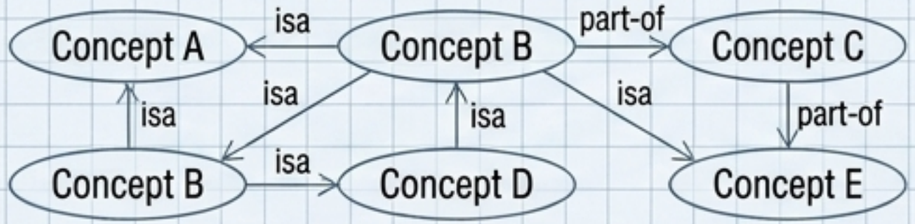
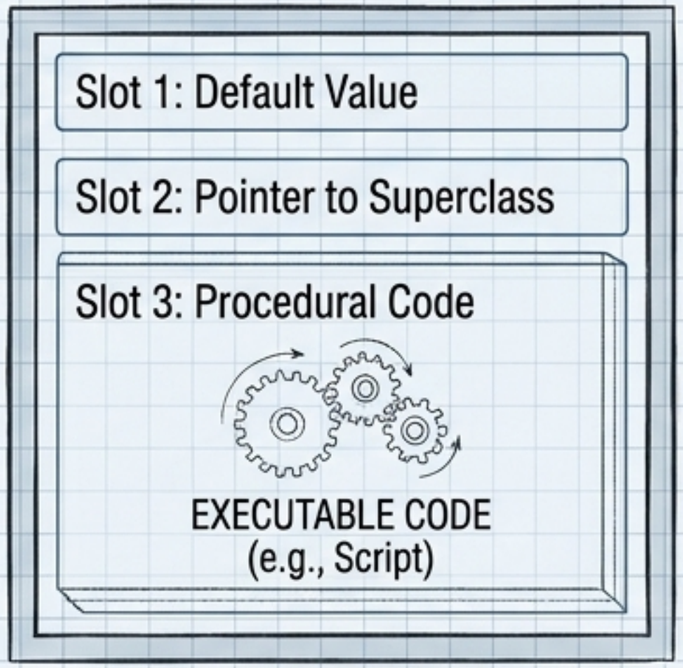
SEMANTIC NETWORKS vs. FRAMES

SEMANTIC NETWORK



FRAMES



Diagnostic Prompt 09

As representational models scale in complexity, the distinction between Semantic Networks and Frames blurs. According to Minsky's foundational concepts (e.g., Scripts), what fundamentally elevates a Frame system over a basic Semantic Network?

- A. The ability to use pointers to indicate superclasses.
- B. The reliance on simple ASCII formatting for rapid data retrieval.
- C. The integration of nodes and arcs to capture relationships.
- ✓ D. The ability to attach executable procedural instructions (code) to slots.

Architect's Note

Toggle D is correct. While both use inheritance and default values, Frames become vastly more powerful when their slots contain procedural knowledge—instructions for computing things, adapting expectations (Scripts), or handling situations when expectations fail.

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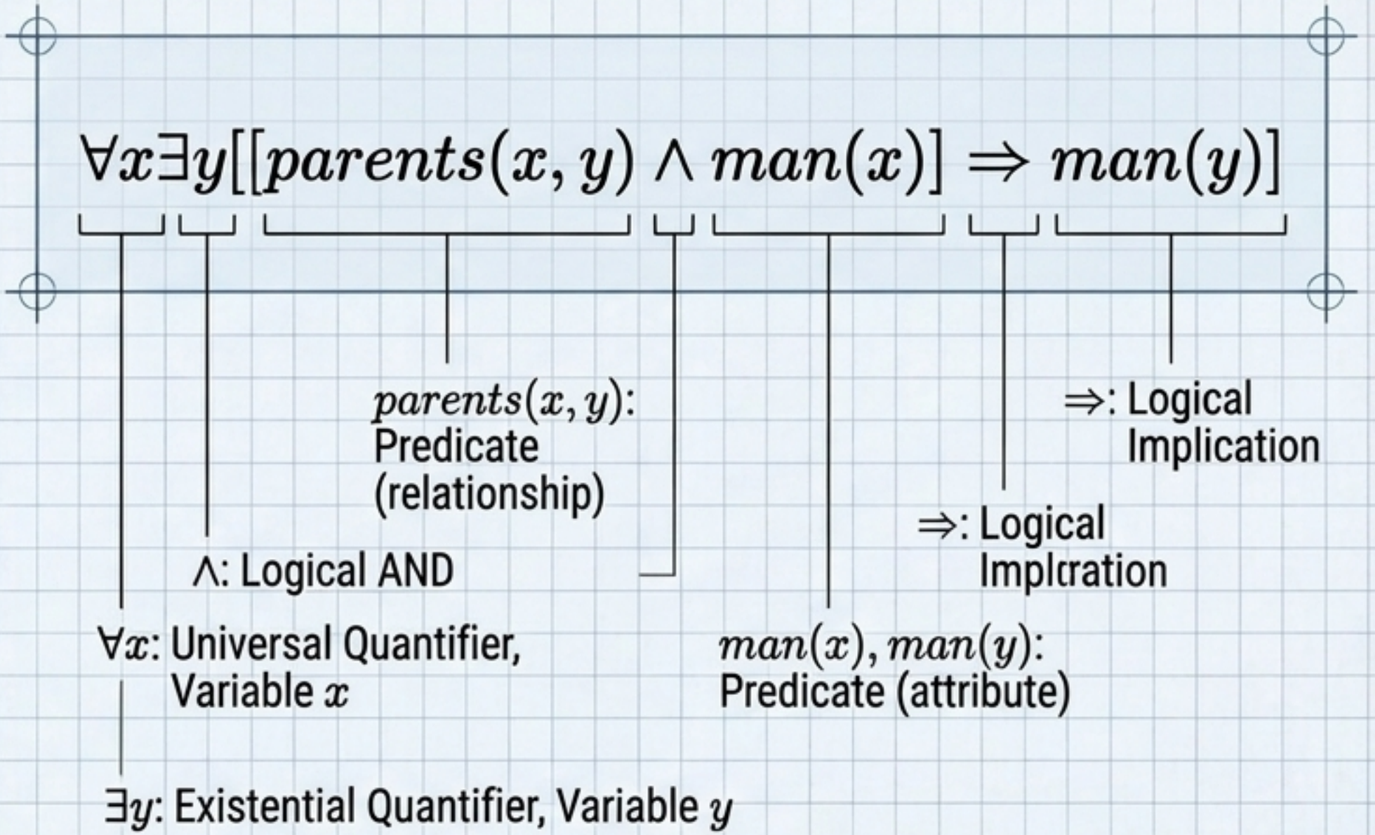
FIRST-ORDER LOGIC: EXPRESSIVENESS & COMPLEXITY

Diagnostic Prompt 10

First-Order Logic (FOL) represents knowledge via constants, variables, predicates, and logical symbols. What is the primary trade-off of selecting FOL as a representation scheme?

- [x] A. High expressiveness and precise semantics, but potentially low inferential efficiency (computational complexity).
- [ ] B. Infinite acquisitional efficiency, but zero representational adequacy.
- [ ] C. Excellent for handling vagueness and uncertainty, but lacks formal syntax.
- [ ] D. Highly modular, but relies exclusively on "if-then" procedural execution.



Architect's Note

Toggle A is correct. A logical framework provides precise syntax, precise semantics, and rigorous reasoning mechanisms (like deduction). However, this high expressiveness comes at the cost of complexity and time consumption, severely impacting inferential efficiency.

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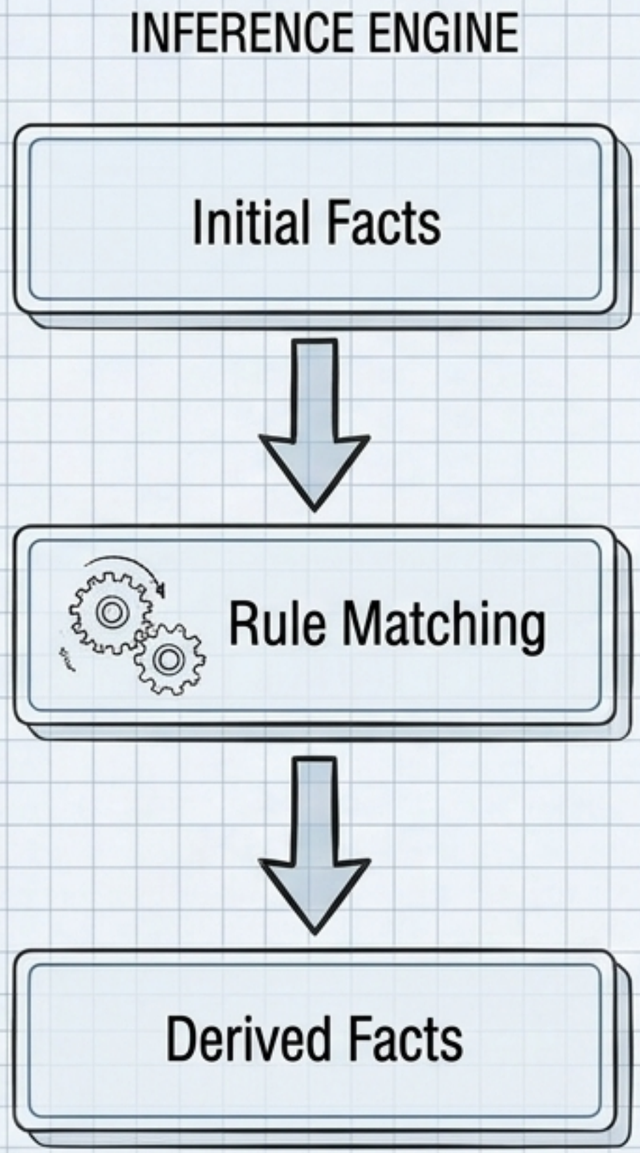
RULE-BASED INFERENCE ENGINE & DIAGNOSTIC PROMPT 11



Diagnostic Prompt 11

Given facts: Raining, Outside(John), ¬HasUmbrella(John).

Given Rules:
R2: IF Raining ∧ Outside(x) ∧ ¬HasUmbrella(x) **THEN** GetWet(x)
R3: IF GetsWet(x) **THEN** CatchCold(x)

Through rule-based inference, what is the final state of the Knowledge Base?

- ○ A. The system loops indefinitely between R2 and R3.
- ✓ **B.** 5 Facts: The initial three, plus GetsWet(John) and CatchCold(John).
- ○ C. 4 Facts: The initial three, plus GetSunBurnt(John).
- ○ D. 3 Facts: The inference engine fails due to the negation operator.

Architect's Note

Toggle B is correct. The rule interpreter matches the conditions of R2 to deduce GetsWet(John). This new fact is added to the database. The interpreter then matches GetsWet(John) to R3, allowing it to infer CatchCold(John). The derivation of new facts from given facts is called forward inference.

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KNOWLEDGE REPRESENTATION PARADIGMS: DIAGNOSTIC MATRIX

Paradigm	Core Mechanism
Logical	Uses formal logic expressions (e.g., Predicate Calculus).
Procedural	Graph nodes representing objects, arcs representing relations.
Network	Instructions for solving problems (e.g., if...then...else).
Structured	Complex data structures with slots, values, and procedures (e.g., Frames).

Diagnostic Prompt 12

Which paradigm in the matrix has been incorrectly matched with its Core Mechanism?

- ○ A. Logical
- ✓ B. Procedural (Procedural is matched with Network's definition).
- ○ C. Network
- ○ D. Structured.

Diagnostic Prompt 13

Which paradigm serves as a direct, engineered extension of the Network scheme, adding procedural power?

- ○ A. Logical
- ○ B. Procedural
- ○ C. Network
- ✓ D. Structured (Structured/Frames).

Architect's Note

Network schemes (Semantic Nets) capture knowledge as graphs. Procedural schemes represent knowledge as if-then instruction sets. Structured schemes (Frames) extend networks by assigning deep slot structures and procedures to each node.

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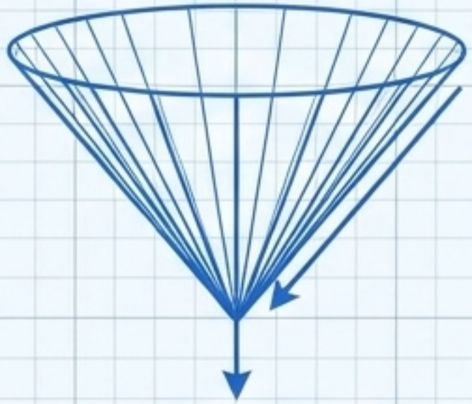
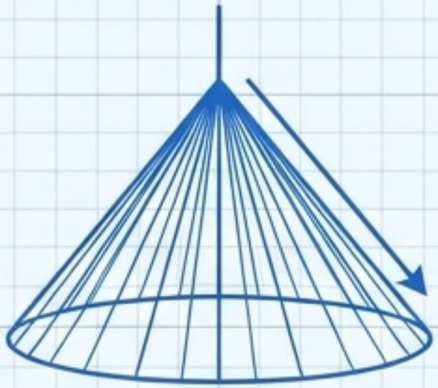
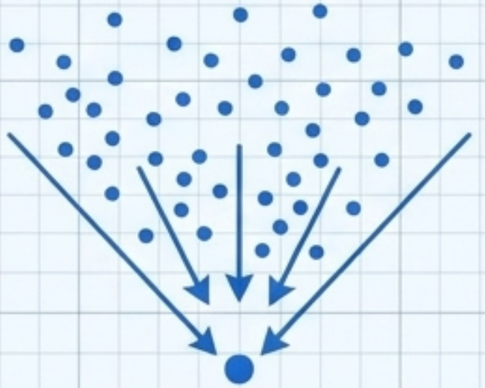
REASONING DIAGRAMS & DIAGNOSTIC PROMPT 14



General to Specific



Particular to General



Plausible Inference

Diagnostic Prompt 14

An AI diagnoses a patient. It knows X implies Y. It observes Y is true. Using probability, it provides a "plausible inference" that X is true.

Which form of reasoning is the system utilising?

- A. Deductive Reasoning
- B. Inductive Reasoning
- C. Abductive Reasoning
- D. Sub-symbolic Reasoning

Architect's Note

Toggle C is correct. Abduction yields the most probable inference based on plausible conditions (If Y is true and X implies Y, then X is true).

Deduction forces a necessary conclusion from general premises (General to Specific). Induction builds a general conclusion from specific instances (Paricular to General).

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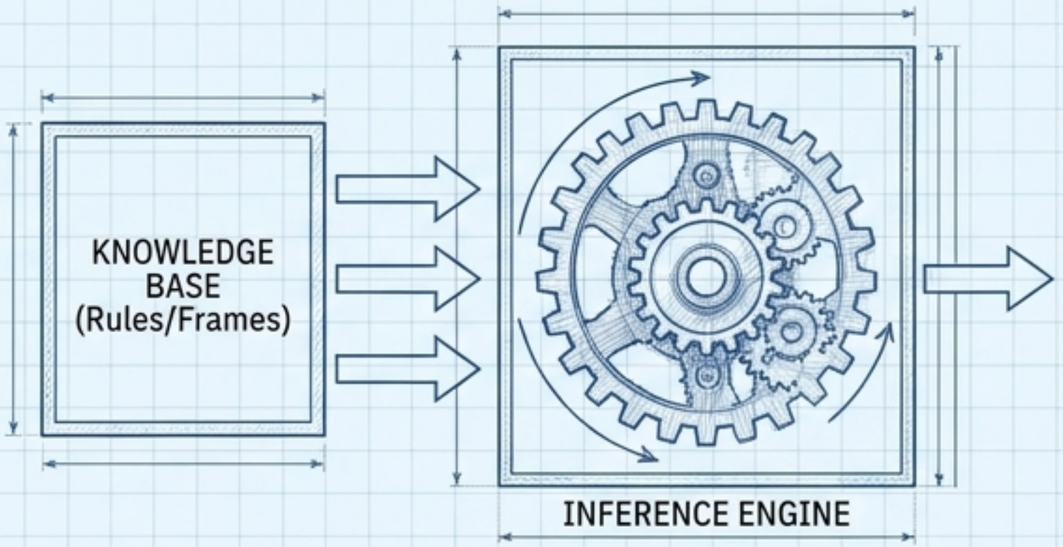

DIAGNOSTIC PROMPT 15 (Synthesis)

To model this ski resort, a team undertakes Knowledge Acquisition, Validation, Representation, and setup of Inferencing algorithms. They compile rules and frames into a Knowledge Base, and activate an Inference Engine. Which term describes the human modeling process, and which describes the final artificial architecture?

- A. Process: Knowledge Management (KM); Architecture: Database System.
- B. Process: Knowledge Engineering (KE); Architecture: Knowledge Based System (KBS).
- C. Process: Semantic Networking; Architecture: First-Order Logic.
- D. Process: Knowledge Based System (KBS); Architecture: Knowledge Engineering (KE).

ARCHITECT'S NOTE

Toggle B is correct. Knowledge Engineering (KE) is the difficult human discipline of modeling the 'right' abstraction (Acquisition, Validation, Representation, Inferencing, Explanation). The Knowledge Based System (KBS) is the resulting machine architecture, consisting centrally of the Knowledge Base and the Inference Engine.

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